HW1: mRNA to protein plots

General Instructions

1. Answer all questions in this R Markdown document.
2. Leave all chunks visible so that we can check your code.
3. Click the **Knit** button to generate a PDF document that includes content, chunk and output.
4. Upload PDF file to Canvas.
5. All questions are worth 1 point.
6. You can find great documentation with examples of most function you will use in this assignment in the following link: https://dplyr.tidyverse.org/reference/.

Case Study: relation between mRNA to protein levels

Despite expectations of a high correlation between mRNA and protein levels, experimental results have shown very low correlation values.

In class, we have been working with a data set from a research group that wants to investigate the relation between mRNA and protein levels.

In this assignment, you will practice your data wrangling skills and create reports with Rmd.

1. Loading the data

Question 1.1

Use `read_csv` to load the mRNA and protein data. They are stored in the data folder. Store them in variables with meaningful names. Make sure all the column names are appropriate.

```
# Your answer goes here.

### BEGIN SOLUTION
prot_raw <-
  read_csv("data/proteinUN.csv") %>%
  rename(gene = ...1)

mrna_raw <-
  read_csv("data/geneUN.csv") %>%
  rename(gene = ...1)
### END SOLUTION
```

Question 1.2.

How many tissues (from organs) have been analyzed?

```
# Your answer goes here

### BEGIN SOLUTION
ncol(prot_raw) - 1
```

```
## [1] 12
```

```r
ncol(mrna_raw) - 1
```

```
## [1] 12
```

```r
### END SOLUTION
```

Question 1.3

Are the dimensions of the mRNA and Proteins data sets the same?

```r
# Your answer goes here

### BEGIN SOLUTION
dim(prot_raw)
```

```
## [1] 6104   13
```

```r
dim(mrna_raw)
```

```
## [1] 6104   13
```

```r
### END SOLUTION
```

2. Getting data ready - Tyding the data

Question 2.1

Gather the tissue columns in the protein data frame and put them in a column called `tissues`, and the values in a column called `protein`. Do the same thing for the mRNA data frame, but name the value column as `mrna`. Remember to save the results in different objects than the original data. (Hint: take a look at the function `pivot_longer`).

```r
# Your answer goes here

### BEGIN SOLUTION
prot_longer <-
  prot_raw %>%
  pivot_longer(!gene, names_to = 'tissues', values_to = "prot")

mrna_longer <-
  mrna_raw %>%
  pivot_longer(!gene, names_to = 'tissues', values_to = "mrna")

head(prot_longer)
```

```
## # A tibble: 6 x 3
##   gene            tissues          prot
##   <chr>           <chr>           <dbl>
## 1 ENSG00000000003 uterus   NA
## 2 ENSG00000000003 kidney   NA
## 3 ENSG00000000003 testis   NA
## 4 ENSG00000000003 pancreas NA
## 5 ENSG00000000003 stomach   0.000000916
## 6 ENSG00000000003 prostate NA
```

```r
head(mrna_longer)
```

```
## # A tibble: 6 x 3
```

```
##   gene            tissues       mrna
##   <chr>           <chr>        <dbl>
## 1 ENSG00000000003 uterus   0.0000250
## 2 ENSG00000000003 kidney   0.0000444
## 3 ENSG00000000003 testis   0.0000579
## 4 ENSG00000000003 pancreas 0.0000143
## 5 ENSG00000000003 stomach  0.0000201
## 6 ENSG00000000003 prostate 0.0000304
```

```
### END SOLUTION
```

Question 2.2:

Join the datasets you obtained in Question 2.1 and save the result in an object called `mrna_prot`.

```
# Your answer goes here

### BEGIN SOLUTION
mrna_prot <-
  prot_longer %>%
  inner_join(mrna_longer)
```

```
## Joining, by = c("gene", "tissues")
```

```
head(mrna_prot)
```

```
## # A tibble: 6 x 4
##   gene            tissues          prot      mrna
##   <chr>           <chr>           <dbl>     <dbl>
## 1 ENSG00000000003 uterus   NA           0.0000250
## 2 ENSG00000000003 kidney   NA           0.0000444
## 3 ENSG00000000003 testis   NA           0.0000579
## 4 ENSG00000000003 pancreas NA           0.0000143
## 5 ENSG00000000003 stomach   0.000000916 0.0000201
## 6 ENSG00000000003 prostate NA           0.0000304
```

```
### END SOLUTION
```

Question 2.3:

Use the `group_by` and `summarise` functions to obtain the *mean*, *std. dev.* of the protein and mrna levels per gene, as well as the number of complete observations (i.e., the number of tissues that have both, mrna and protein levels) in columns named: `mean_prot`, `sd_prot`, `mean_mrna`, `sd_mrna`, and `n_comp_obs`. Remove the `NA` in the calculation. Save the result in an object named `summary_mrna_prot`.

```
# Your answer goes here

### BEGIN SOLUTION
summary_mrna_prot <-
  mrna_prot %>%
  group_by(gene) %>%
  summarise(mean_prot = mean(prot, na.rm = TRUE),
            sd_prot = sd(prot, na.rm = TRUE),
            mean_mrna = mean(mrna, na.rm = TRUE),
            sd_mrna = sd(mrna, na.rm = TRUE),
            n_comp_obs = sum(!is.na(prot) & !is.na(mrna)))

head(summary_mrna_prot)
```

```
## # A tibble: 6 x 6
##   gene             mean_prot    sd_prot  mean_mrna    sd_mrna n_comp_obs
##   <chr>                <dbl>      <dbl>      <dbl>      <dbl>      <int>
## 1 ENSG00000000003 0.00000497 0.00000574 0.0000345  0.0000172           2
## 2 ENSG00000000419 0.0000202  0.0000151  0.0000346  0.00000843         12
## 3 ENSG00000000457 0.00000262 0.00000327 0.00000722 0.00000118          5
## 4 ENSG00000000971 0.000197   0.000137   0.0000279  0.0000176          12
## 5 ENSG00000001036 0.0000152  0.0000165  0.0000335  0.0000126           7
## 6 ENSG00000001084 0.0000444  0.0000427  0.0000201  0.00000959         12
```

```
### END SOLUTION
```

Question 2.4

Using the `filter` and `nrow` functions, obtain how many genes have mRNA and protein levels in *all* 12 tissues.

```
# Your answer goes here

### BEGIN SOLUTION
summary_mrna_prot %>%
  filter(n_comp_obs == 12) %>%
  nrow()
```

```
## [1] 1392
```

```
### END SOLUTION
```

Question 2.5

Let's bring the information in the `summary_mrna_prot` into our `tibble` `mrna_prot` with the data. (Hint: see `left_join`.) Remember to save the data.

```
# Your answer goes here

### BEGIN SOLUTION
mrna_prot <-
  mrna_prot %>%
  left_join(summary_mrna_prot)
```

```
## Joining, by = "gene"
```

```
head(mrna_prot, 20)
```

```
## # A tibble: 20 x 9
##    gene  tissues     prot    mrna mean_prot sd_prot mean_mrna sd_mrna n_comp_obs
##    <chr> <chr>      <dbl>   <dbl>     <dbl>   <dbl>     <dbl>   <dbl>      <int>
##  1 ENSG~ uterus  NA       2.50e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
##  2 ENSG~ kidney  NA       4.44e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
##  3 ENSG~ testis  NA       5.79e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
##  4 ENSG~ pancre~ NA       1.43e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
##  5 ENSG~ stomach  9.16e-7 2.01e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
##  6 ENSG~ prosta~ NA       3.04e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
##  7 ENSG~ ovary   NA       5.60e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
##  8 ENSG~ thyroi~ NA       3.04e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
##  9 ENSG~ adrena~ NA       2.16e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
## 10 ENSG~ saliva~  9.03e-6 5.58e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
## 11 ENSG~ spleen  NA       9.90e-6   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
```

```
## 12 ENSG~ esopha~ NA       4.76e-5   4.97e-6 5.74e-6 0.0000345 1.72e-5          2
## 13 ENSG~ uterus   9.97e-6 3.44e-5   2.02e-5 1.51e-5 0.0000346 8.43e-6         12
## 14 ENSG~ kidney   1.89e-5 3.43e-5   2.02e-5 1.51e-5 0.0000346 8.43e-6         12
## 15 ENSG~ testis   4.26e-5 3.80e-5   2.02e-5 1.51e-5 0.0000346 8.43e-6         12
## 16 ENSG~ pancre~  1.58e-6 2.11e-5   2.02e-5 1.51e-5 0.0000346 8.43e-6         12
## 17 ENSG~ stomach  3.28e-5 2.96e-5   2.02e-5 1.51e-5 0.0000346 8.43e-6         12
## 18 ENSG~ prosta~  2.12e-5 3.40e-5   2.02e-5 1.51e-5 0.0000346 8.43e-6         12
## 19 ENSG~ ovary    3.92e-5 3.17e-5   2.02e-5 1.51e-5 0.0000346 8.43e-6         12
## 20 ENSG~ thyroi~  3.93e-6 4.54e-5   2.02e-5 1.51e-5 0.0000346 8.43e-6         12
```

```r
### END SOLUTION
```

Question 2.6

Is the missing information dependent on the organ? In other words, do some organs' tissues have more missing data than others? To answer this, let's calculate the proportion of genes with `prot` and/or `mrna` missing for each tissue.

If the genes in our sample were randomly selected, we could even calculate the confidence interval for the proportions. Although the assumption of randomly selected genes is not true, for the sake of discussion, let's still include an approximate 95% confidence interval $\hat{p} \pm 1.96 \times \sqrt{\hat{p}(1-\hat{p})/n}$.

Save it in an object named `prop_missing_mrna_prot`. (*Hint: again, you'll need* `group_by` *and* `summarise`).

```r
# Your answer goes here

### BEGIN SOLUTION
prop_missing_mrna_prot <-
  mrna_prot %>%
  group_by(tissues) %>%
  summarise(p_hat = sum(is.na(prot) | is.na(mrna)) / n(),
            lower_ci = p_hat - 1.96 * sqrt(p_hat*(1-p_hat)/n()),
            upper_ci = p_hat + 1.96 * sqrt(p_hat*(1-p_hat)/n()))
prop_missing_mrna_prot
```

```
## # A tibble: 12 x 4
##    tissues        p_hat lower_ci upper_ci
##    <chr>          <dbl>    <dbl>    <dbl>
##  1 adrenal.gland  0.365    0.353    0.377
##  2 esophagus      0.404    0.391    0.416
##  3 kidney         0.486    0.473    0.498
##  4 ovary          0.377    0.365    0.389
##  5 pancreas       0.503    0.490    0.515
##  6 prostate       0.401    0.388    0.413
##  7 salivary.gland 0.305    0.294    0.317
##  8 spleen         0.470    0.458    0.483
##  9 stomach        0.353    0.341    0.365
## 10 testis         0.335    0.323    0.347
## 11 thyroid.gland  0.568    0.556    0.581
## 12 uterus         0.545    0.533    0.558
```

```r
### END SOLUTION
```

Question 2.7

The summary you stored in `prop_missing_mrna_prot` is very useful but hard to read and compare all the numbers. Let's use `ggplot2` to create a visualization that will help us to interpret the table.

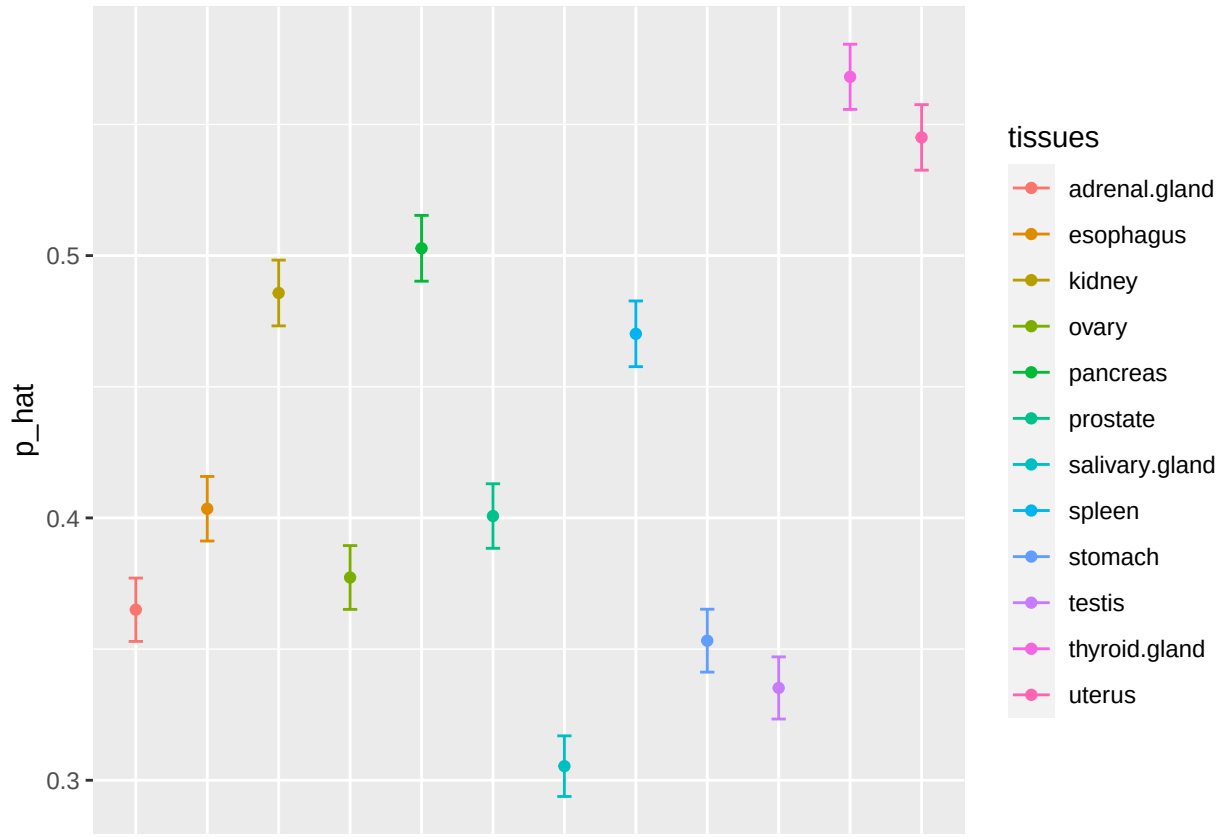
We will walk you through creating the visualization in the scaffolding below.

(Note: remember that our gene sample might not be random, so the validity of these CI are questionable).

```
prop_missing_mrna_prot %>%
  ggplot(aes(x = ..., # We want one bar per tissue
             y = ..., # the estimate proportion
             colour = ... # let's give different colors for each tissue
             )
         ) +
  geom_...() + # replace the three dots with the
  geom_errorbar(aes(ymin = ..., # the lower limit of the confidence interval
                    ymax = ..., # the upper limit of the confidence interval
                    width = .2 # this value controls the size of horizontal bar (try changing it))
                    ))  +
  theme(axis.title.x = element_blank(), # You don't need to worry about this code
        axis.text.x = element_blank(),  # it is only to remove the x-axis.
        axis.ticks.x = element_blank())
```

```
### BEGIN SOLUTION

prop_missing_mrna_prot %>%
  ggplot(aes(x = tissues, # We want one bar per tissue
             y = p_hat, # the estimate proportion
             colour = tissues # let's give different colors for each tissue
             )
         ) +
  geom_point() + # replace the three dots with the
  geom_errorbar(aes(ymin = lower_ci, # the lower limit of the confidence interval
                    ymax = upper_ci, # the upper limit of the confidence interval
                    width = .2 # this value controls the size of horizontal bar (try changing it))
                    ))  +
  theme(axis.title.x = element_blank(), # You don't need to worry about this code
        axis.text.x = element_blank(),  # it is only to remove the x-axis.
        axis.ticks.x = element_blank())
```




```
### END SOLUTION
```

3. Genes and Sampling

Question 3.1 Select four different genes with no missing data, using the `select`, `unique`, `filter`, and `slice_sample` functions. To make sure we can reproduce your plot, we will set the seed to 450. Save the result in an object named `sample_genes`. Make sure your code cannot select the same gene more than once.

```
set.seed(450) # do not erase this.

# Your answer goes here.

### BEGIN SOLUTION
sample_genes <-
  mrna_prot %>%
  filter(n_comp_obs == 12) %>%
  select(gene) %>%
  unique() %>%
  slice_sample(n = 4) %>%
  pull()

sample_genes
```

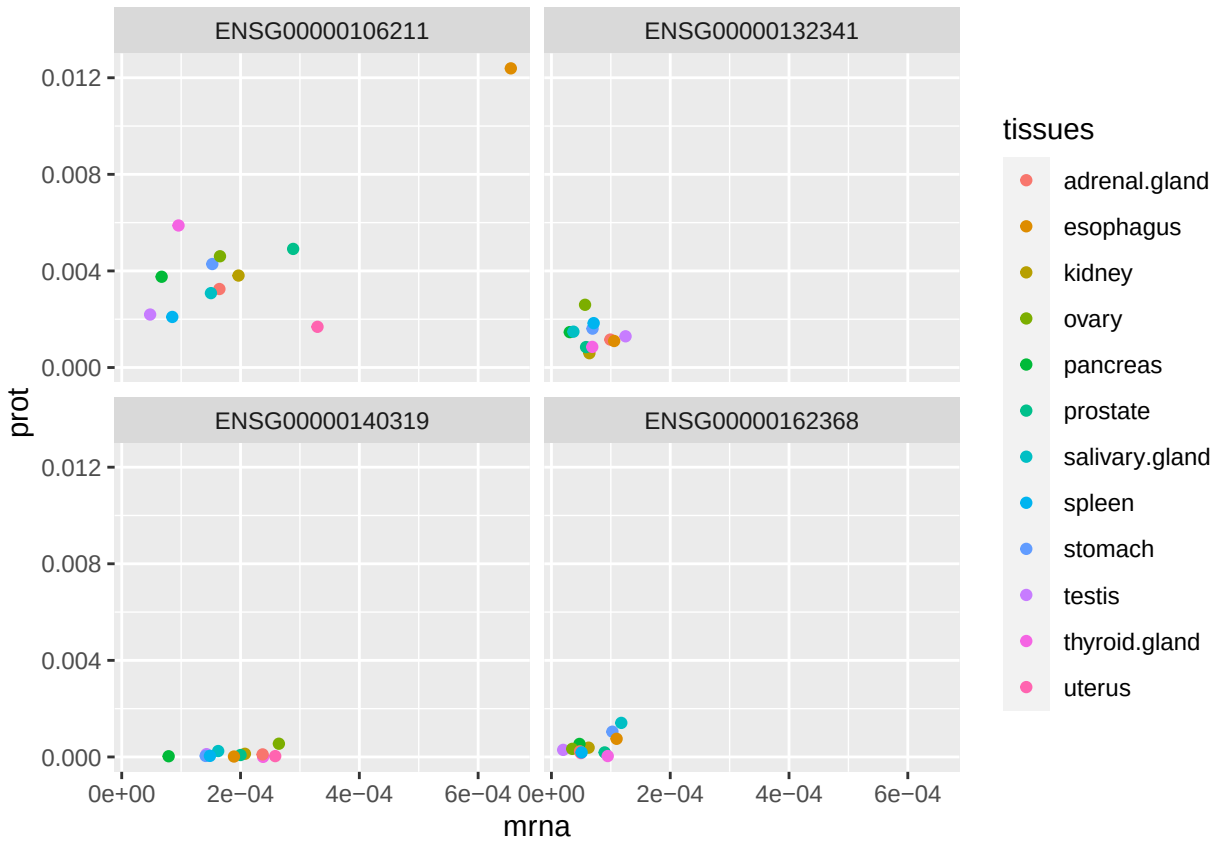
```
## [1] "ENSG00000162368" "ENSG00000132341" "ENSG00000106211" "ENSG00000140319"
```

```
### END SOLUTION
```

Question 3.2 Create a scatter plot that illustrates the relation between mRNA and protein for the selected genes; colour the points by tissue type. Use the `facet_wrap` function to separate the plots for each gene.

```
# Your answer goes here.

### BEGIN SOLUTION
mrna_prot %>%
  filter(gene %in% sample_genes) %>%
  ggplot(aes(mrna, prot, color = tissues)) +
  geom_point() +
  facet_wrap(~gene)
```




```
### END SOLUTION
```